Characterize efficient learning when perceived and true signal tail exponents coincide

ChatGPT (`gpt-5.6-sol`)*

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Abstract

This paper gives a complete solution to the efficient-learning problem when the true and perceived private-belief distributions have the same lower-tail exponent. The common exponent alone does not determine whether the expected total number of erroneous actions is finite. For arbitrary continuous distributions satisfying two-sided power bounds, we derive an exact necessary-and-sufficient deterministic renewal-series criterion. If the true and perceived mixture distributions satisfy

$$G(x) \sim cx^{\alpha}, \qquad \widetilde{G}(x) \sim \widetilde{c}x^{\alpha},$$

then learning is efficient when $c > \alpha\widetilde{c}$ and inefficient when $c < \alpha\widetilde{c}$. At the critical ratio $c = \alpha\widetilde{c}$, first-order tail coefficients remain insufficient: a logarithmic perturbation with parameter γ is efficient exactly when $\gamma > 1$. Explicit coherent equal-exponent examples are given on both sides of the threshold and at the harmonic boundary. The equal-exponent case is thus completely resolved under the stated hypotheses; there is no remaining gap.

1 The open problem

This paper addresses an open problem posed by Itai Arieli, Yakov Babichenkoyako, Stephan Müller, Farzad Pourbabaee, Omer Tamuz in *The Hazards and Benefits of Condescension in Social Learning* [1] (Economics and Computation (EC), 2023), Section 3 Results, page 7 (discussion immediately after Theorem 1).. The website catalogue entry is problem 5 (`W4383533356_p0`) of the Open Problems in Operations Research collection.

> *Assume the sequential social learning model above with symmetry, continuity, and tail-regularity, and suppose the true and perceived tail exponents coincide, $\tilde{\alpha} = \alpha$. Determine necessary and sufficient conditions, stated in terms of the distributions (F_ℓ, F_h) and $(\tilde{F}_\ell, \tilde{F}_h)$ beyond the common tail exponent, for efficient learning:*
>
> $$\mathbb{E}[W] < \infty,$$
>
> *where $W = \#\{n : a_n \neq \theta\}$ under the equilibrium induced by the misspecified beliefs $(\tilde{F}_\ell, \tilde{F}_h)$ and the true signal process (F_ℓ, F_h).*

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260824_024352_445006f1, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

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2 Introduction and model

There are two states, denoted by h and ℓ. The initial public probability of state h is

$$\pi \in (0,1),$$

and the true state is drawn according to this prior. Agents act sequentially. Conditional on the state, their private signals are independent and identically distributed.

It is convenient to represent a private signal by the posterior probability

$$Q \in [0,1]$$

that it assigns to state h when the reference prior on the two states is $1/2$. Let F_{h} and F_ℓ be the true conditional distribution functions of Q, and let $\widetilde{F}_{\mathsf{h}}$ and $\widetilde{F}_\ell$ be the conditional distribution functions perceived by the agents. Define the true and perceived mixture distributions by

$$G := \frac{F_\ell + F_{\mathsf{h}}}{2}, \qquad \widetilde{G} := \frac{\widetilde{F}_\ell + \widetilde{F}_{\mathsf{h}}}{2}.$$

We impose the following hypotheses.

1. The distributions G and $\widetilde{G}$ are continuous and symmetric about $1/2$.

2. Both the true and the perceived private-belief models are Bayes coherent. Equivalently, as identities of finite measures on $[0,1]$,

$$dF_{\mathsf{h}}(q) = 2q\,dG(q), \qquad dF_\ell(q) = 2(1-q)\,dG(q), \tag{1}$$

and

$$d\widetilde{F}_{\mathsf{h}}(q) = 2q\,d\widetilde{G}(q), \qquad d\widetilde{F}_\ell(q) = 2(1-q)\,d\widetilde{G}(q). \tag{2}$$

3. Agents observe all preceding actions and update the public belief using the perceived conditional distributions. If the current public probability of h is p, an agent chooses action h precisely when

$$p + Q \geq 1. \tag{3}$$

   Continuity makes the treatment of equality immaterial.

4. For some common exponent $\alpha > 0$,

$$G(x) = \Theta(x^\alpha), \qquad \widetilde{G}(x) = \Theta(x^\alpha) \qquad (x \downarrow 0). \tag{4}$$

   Thus there are positive constants $m, M, \widetilde{m}, \widetilde{M}$ and $\delta > 0$ such that, for every $x \in (0,\delta]$,

$$mx^\alpha \leq G(x) \leq Mx^\alpha, \qquad \widetilde{m}x^\alpha \leq \widetilde{G}(x) \leq \widetilde{M}x^\alpha.$$

The calibration identities follow directly from the interpretation of Q as a posterior. Indeed, for every Borel set $E \subseteq [0,1]$, under the reference prior $1/2$,

$$\frac{1}{2}F_{\mathsf{h}}(E) = \mathbb{P}(\mathsf{h},\, Q \in E) = \int_E q\, dG(q),$$

and similarly

$$\frac{1}{2}F_{\ell}(E) = \int_E (1-q)\, dG(q).$$

The same argument applies to the perceived model. All integrals below are consequently Stieltjes integrals; no density or absolute-continuity assumption is needed.

An action is erroneous if it differs from the true state. Let

$$W := \sum_{t=1}^{\infty} \mathbf{1}\{\text{the action of agent } t \text{ differs from the true state}\}.$$

We call learning *efficient* if

$$\mathbb{E}[W] < \infty.$$

The purpose of this paper is to characterize efficiency under (4). The result is a complete solution, rather than partial progress: an exact criterion is obtained for arbitrary continuous two-sided power tails, and simpler classifications are obtained when additional tail expansions exist.

3 The two public-belief update maps

For $x \in (0, 1/2]$, define

$$A(x) := F_{\ell}(x) = 2\int_0^x (1-u)\, dG(u), \tag{5}$$

$$B(x) := F_{\mathsf{h}}(x) = 2\int_0^x u\, dG(u). \tag{6}$$

Define $\widetilde{A}$ and $\widetilde{B}$ analogously using $\widetilde{G}$.

At any history, let x be the subjective probability assigned to the state opposite the current public majority. If the public belief is tied, either state may be designated as the majority; symmetry makes the resulting analysis identical.

Suppose first that the current majority is h, so its public probability is $1-x$. By (3), an agent agrees with the majority exactly when $Q \geq x$. Under the perceived model, the probabilities of agreement conditional on states ℓ and h are, respectively,

$$1 - \widetilde{A}(x) \quad \text{and} \quad 1 - \widetilde{B}(x).$$

Bayes' rule therefore sends the minority probability x to

$$S(x) := \frac{x\big(1-\widetilde{A}(x)\big)}{x\big(1-\widetilde{A}(x)\big) + (1-x)\big(1-\widetilde{B}(x)\big)} \tag{7}$$

after an action agreeing with the current majority.

If the action instead opposes the current majority, its perceived probabilities in states ℓ and h are $\widetilde{A}(x)$ and $\widetilde{B}(x)$. The action reverses the public majority, and the probability assigned to the new minority state is

$$R(x) := \frac{(1-x)\widetilde{B}(x)}{x\widetilde{A}(x) + (1-x)\widetilde{B}(x)}. \tag{8}$$

When the current majority is ℓ, symmetry gives the same two maps.

To verify that an opposing action really reverses the majority, observe that, for every $u \le x$,

$$(1-x)u \le x(1-u).$$

Integration against $2\,d\widetilde{G}(u)$ gives

$$(1-x)\widetilde{B}(x) \le x\widetilde{A}(x),$$

and hence $R(x) \le 1/2$.

If the current majority is correct, the true probability that the next action is an erroneous reversal is $B(x)$. If the current majority is wrong, the true probability that the next action is a correct rescuing reversal is $A(x)$.

A direct calculation from (7) gives the useful identity

$$x - S(x) = \frac{x(1-x)\big(\widetilde{A}(x) - \widetilde{B}(x)\big)}{1 - x\widetilde{A}(x) - (1-x)\widetilde{B}(x)}. \tag{9}$$

4 Tail estimates and deterministic confidence dynamics

Lemma 1 (Conditional-tail orders)**.** *Under* (4)*,*

$$A(x) = \Theta(x^{\alpha}), \qquad B(x) = \Theta(x^{\alpha+1}) \qquad (x \downarrow 0), \tag{10}$$

and the same conclusions hold for $\widetilde{A}$ *and* $\widetilde{B}$*.*

Proof. For $x \le 1/2$,

$$2(1-x)G(x) \le A(x) \le 2G(x),$$

so $A(x) = \Theta(x^{\alpha})$. Also,

$$B(x) \le 2xG(x) \le 2Mx^{\alpha+1}.$$

For the lower bound, choose $\lambda \in (0,1)$ so small that

$$M\lambda^{\alpha} \le \frac{m}{2}.$$

Continuity of G and the two-sided tail bounds give

$$\begin{aligned} B(x) &\ge 2\lambda x\big(G(x) - G(\lambda x)\big) \\ &\ge 2\lambda x\big(m - M\lambda^{\alpha}\big)x^{\alpha} \\ &\ge \lambda m x^{\alpha+1} \end{aligned}$$

for all sufficiently small x. This proves the assertion for the true model. The proof for the perceived model is identical. $\square$

Since
$$\widetilde{A}(x) - \widetilde{B}(x) = 2\widetilde{G}(x) - 2\widetilde{B}(x) = \Theta(x^\alpha),$$
and the denominator in (9) tends to 1, it follows that
$$x - S(x) = \Theta(x^{\alpha+1}). \tag{11}$$

For $z \in (0, 1/2]$, write
$$x_n(z) := S^n(z), \qquad n \geq 0. \tag{12}$$

Lemma 2 (Power-law decay of deterministic orbits)**.** *For every compact set $J \subset (0, 1/2]$, there are constants $a_J, b_J > 0$ such that*
$$a_J(n+1)^{-1/\alpha} \leq x_n(z) \leq b_J(n+1)^{-1/\alpha} \tag{13}$$
for all $z \in J$ and all $n \geq 0$.

If, more precisely,
$$S(x) = x - ax^{\alpha+1}(1 + o(1)) \qquad (x \downarrow 0) \tag{14}$$
for some $a > 0$, then
$$x_n(z)^\alpha \sim \frac{1}{a\alpha n}. \tag{15}$$
The convergence in (15) is uniform for z in a fixed compact subset of $(0, 1/2]$.

Proof. For $x \in (0, 1/2]$,
$$\widetilde{A}(x) - \widetilde{B}(x) = 2\int_0^x (1 - 2u)\, d\widetilde{G}(u) > 0,$$
because the lower-tail assumption supplies positive mass below x. Thus $S(x) < x$. Continuity and strict inequality imply that, uniformly over a compact set of starting points, the orbit enters any prescribed sufficiently small interval in finitely many steps.

By (11), there are $d_1, d_2, \delta_1 > 0$ such that
$$d_1 x^{\alpha+1} \leq x - S(x) \leq d_2 x^{\alpha+1}$$
whenever $x \leq \delta_1$. Decreasing δ_1 if necessary also ensures $S(x) \geq x/2$.

Put $u_n = x_n^{-\alpha}$. By the mean-value theorem,
$$u_{n+1} - u_n = \alpha \xi_n^{-\alpha-1}(x_n - x_{n+1})$$
for some $\xi_n \in [x_{n+1}, x_n]$. Since $x_n/2 \leq \xi_n \leq x_n$, the increments $u_{n+1} - u_n$ are bounded above and below by positive constants independent of n once the orbit is in the small interval. Summation gives $u_n = \Theta(n)$, uniformly over compact sets of starting points. This proves (13).

Under (14),
$$\frac{x_{n+1}}{x_n} = 1 - ax_n^\alpha(1 + o(1)) \longrightarrow 1.$$
Applying the mean-value formula again gives
$$u_{n+1} - u_n = a\alpha + o(1).$$
Consequently,
$$u_n = u_0 + \sum_{k=0}^{n-1} \big(a\alpha + o(1)\big) = a\alpha n + o(n),$$
which is equivalent to (15). The tail estimates are uniform in x, and compact sets enter the tail region uniformly, proving the final assertion. □

Lemma 3 (Interior reset). *There exists $r_0 > 0$ such that*

$$R(x) \in [r_0, 1/2] \qquad \textit{for every } x \in (0, 1/2]. \tag{16}$$

Proof. The upper bound was proved after (8). For small x, the conditional-tail estimates give

$$(1-x)\widetilde{B}(x) = \Theta(x^{\alpha+1})$$

and

$$x\widetilde{A}(x) + (1-x)\widetilde{B}(x) = \Theta(x^{\alpha+1}).$$

Their ratio is therefore bounded below by a positive constant. On every compact subinterval of $(0, 1/2]$, the numerator and denominator in (8) are continuous and strictly positive. Compactness supplies a positive lower bound there as well. $\square$

Let

$$I := [r_0, 1/2]$$

denote the reset region. From (10) and (13),

$$\sum_{n=0}^{\infty} B(x_n(z)) < \infty, \tag{17}$$

uniformly for $z \in I$, because

$$B(x_n(z)) = O\big((n+1)^{-1-1/\alpha}\big).$$

Moreover, $B(1/2) < 1$. Indeed, symmetry and the lower-tail assumption give positive G-mass above $1/2$, and hence positive F_{h}-mass above $1/2$. Therefore there is a constant $\eta > 0$ such that

$$\prod_{n=0}^{\infty} \big(1 - B(x_n(z))\big) \geq \eta \qquad \text{for all } z \in I. \tag{18}$$

Specifically, with $\beta := B(1/2) < 1$,

$$\log(1-u) \geq -\frac{u}{1-\beta}, \qquad 0 \leq u \leq \beta,$$

and the uniform summability in (17) yields (18). Thus every correct-majority run beginning in the reset region has a uniformly positive probability of continuing forever without another error.

5 The exact wrong-run criterion

We first record two elementary extended-valued identities used below.

Lemma 4 (Tail sums and infinite products). *Let N take values in $\{0, 1, 2, \ldots\} \cup \{\infty\}$. Then*

$$\mathbb{E}[N] = \sum_{n=1}^{\infty} \mathbb{P}(N \geq n), \tag{19}$$

with both sides interpreted in $[0, \infty]$.

Let $0 \leq a_k \leq \beta < 1$, and suppose $\sum_k a_k^2 < \infty$. Then

$$\prod_{k=0}^{n-1} (1 - a_k) \asymp \exp\left(-\sum_{k=0}^{n-1} a_k\right). \tag{20}$$

Proof. Pointwise,

$$N = \sum_{n=1}^{\infty} \mathbf{1}_{\{N \geq n\}},$$

including on the event $\{N = \infty\}$. Tonelli's theorem applies because every summand is nonnegative, and it permits the interchange of expectation and the countable sum even when their common value is infinite. This proves (19).

For $0 \leq u \leq \beta < 1$, Taylor's formula, or direct integration of $1/(1-u)$, gives

$$-u - C_\beta u^2 \leq \log(1-u) \leq -u$$

for a finite constant C_β. Summing this inequality and using $\sum_k a_k^2 < \infty$ proves (20). $\square$

Consider a wrong-majority run beginning with minority confidence z. As long as agents continue to choose the wrong majority action, the confidence sequence is $x_n(z)$. At confidence $x_n(z)$, the probability of a rescuing correct action is $A(x_n(z))$. Hence the probability that the first n actions of the run are all wrong is

$$P_n(z) := \prod_{k=0}^{n-1} \bigl(1 - A(x_k(z))\bigr). \tag{21}$$

If N denotes the number of wrong actions before the first rescuing action, with $N = \infty$ if no rescue occurs, then

$$\mathbb{P}(N \geq n) = P_n(z).$$

The tail-sum identity therefore gives the exact expected length

$$L(z) := \mathbb{E}[N] = \sum_{n=1}^{\infty} P_n(z). \tag{22}$$

By (10) and (13),

$$A(x_k(z)) = \Theta((k+1)^{-1}),$$

so

$$\sum_{k=0}^{\infty} A(x_k(z))^2 < \infty.$$

Also $A(1/2) < 1$, because the true conditional law in state ℓ has positive mass above $1/2$. The product comparison consequently gives

$$P_n(z) \asymp \exp\left(-\sum_{k=0}^{n-1} A(x_k(z))\right). \tag{23}$$

The identity

$$A(x) = 2G(x) - B(x)$$

and (17) show that the partial sums of $B(x_k(z))$ remain bounded. Thus

$$P_n(z) \asymp \exp\left(-2\sum_{k=0}^{n-1} G(x_k(z))\right). \tag{24}$$

We have proved the following exact finer-tail test.

Proposition 1 (Exact criterion for a wrong run)**.** *For every* $z \in (0, 1/2]$,

$$\boxed{L(z) < \infty \quad \Longleftrightarrow \quad \sum_{n=1}^{\infty} \exp\left(-2\sum_{k=0}^{n-1} G(x_k(z))\right) < \infty, \qquad x_{k+1}(z) = S(x_k(z)).} \tag{25}$$

No differentiability, density, or tail-constant assumption is required.

6 The exact global renewal criterion

For $z \in (0, 1/2]$, define

$$p_n(z) := \prod_{k=0}^{n-1} \bigl(1 - A(x_k(z))\bigr), \qquad q_n(z) := \prod_{k=0}^{n-1} \bigl(1 - B(x_k(z))\bigr), \tag{26}$$

$$r_n(z) := p_n(z)A(x_n(z)), \qquad e_n(z) := q_n(z)B(x_n(z)), \tag{27}$$

where empty products equal 1. Thus $r_n(z)$ is the probability that a wrong run is rescued after exactly n preceding wrong actions, while $e_n(z)$ is the probability that a correct run first errs after exactly n preceding correct actions.

Set

$$y_n(z) := R(x_n(z)). \tag{28}$$

The reset lemma ensures $y_n(z) \in I$.

Define the nonnegative extended-valued function

$$g(z) := \sum_{i=0}^{\infty} e_i(z)\bigl(1 + L(y_i(z))\bigr). \tag{29}$$

The term 1 counts the erroneous action that ends a correct run. The term $L(y_i(z))$ counts all subsequent wrong actions before rescue. The rescuing action itself is correct and is not counted.

For a nonnegative function f, define the substochastic renewal operator

$$(Kf)(z) := \sum_{i=0}^{\infty} e_i(z) \sum_{j=0}^{\infty} r_j(y_i(z)) f\bigl(y_j(y_i(z))\bigr). \tag{30}$$

The inner transition occurs only when the wrong-majority excursion is eventually rescued.

Theorem 1 (Exact global criterion)**.** *Let* $C(z)$ *be the expected future number of errors conditional on a correct current public majority with minority confidence* z*, and let* $D(z)$ *be the corresponding expectation conditional on a wrong current majority. Then, as identities in* $[0, \infty]$*,*

$$C(z) = \sum_{m=0}^{\infty} (K^m g)(z), \tag{31}$$

$$D(z) = L(z) + \sum_{j=0}^{\infty} r_j(z)C(y_j(z)). \tag{32}$$

Let

$$x_0 := \min\{\pi, 1 - \pi\}.$$

Then

$$\boxed{\mathbb{E}[W] < \infty \quad \iff \quad (1 - x_0)C(x_0) + x_0 D(x_0) < \infty.} \tag{33}$$

All quantities on the right side are explicit deterministic series determined by (7)*,* (8)*, and* (21)*–*(30)*.*

Proof. Starting from a correct majority, the first possible excursion consists of:

1. i correct actions followed by one erroneous reversal, with probability $e_i(z)$;
2. a wrong-majority run beginning at $y_i(z)$, whose expected number of wrong actions is $L(y_i(z))$;
3. if rescue occurs after exactly j wrong actions, a return to a correct majority with confidence $y_j(y_i(z))$.

Consequently, $g(z)$ is exactly the expected number of errors in the first excursion, without counting any errors after its rescue, and K is exactly the transition kernel for continuation after a completed excursion.

Conditional independence of private signals and stationarity of their conditional distributions imply that, given the new confidence and whether the majority is correct, the future process has the same law as a fresh process with those data. Iterating over successive completed excursions therefore shows that the expected number of errors contributed by the $(m + 1)$-st excursion is $(K^m g)(z)$.

All excursion error counts and all terms in the resulting expansion are nonnegative. Tonelli's theorem for countable sums of nonnegative random variables therefore permits summation over excursions without any prior integrability assumption and remains valid if the answer is $+\infty$. This proves (31). No action is counted twice: the error reversing a correct majority appears in the 1 in (29), the subsequent wrong actions appear in L, and the rescuing action is correct.

Starting from a wrong majority, the initial wrong run contributes $L(z)$. If it is rescued after exactly j wrong actions, which has probability $r_j(z)$, the process continues from a correct majority with confidence $y_j(z)$. This proves (32).

Initially, the public majority is correct with probability

$$\max\{\pi, 1 - \pi\} = 1 - x_0$$

and wrong with probability x_0. At $\pi = 1/2$, either tie-breaking designation gives the same two probabilities by symmetry. Conditioning on initial correctness proves

$$\mathbb{E}[W] = (1 - x_0)C(x_0) + x_0 D(x_0)$$

as an extended-valued identity, and hence proves (33). □

The contraction needed to use this criterion is supplied by (18). Indeed,

$$(K\mathbf{1})(z) \le \sum_{i=0}^{\infty} e_i(z) = 1 - \prod_{k=0}^{\infty} \big(1 - B(x_k(z))\big) \le 1 - \eta \tag{34}$$

for $z \in I$. Thus, if

$$\sup_{z \in I} L(z) < \infty,$$

then g is bounded on the reset region and the series (31) is dominated there by a convergent geometric series. If also $L(x_0) < \infty$, then $\mathbb{E}[W] < \infty$. Conversely, if $L(x_0) = \infty$, then $D(x_0) = \infty$, and since $x_0 > 0$, the global expectation is infinite.

7 Classification when tail constants exist

Assume now that

$$G(x) \sim cx^{\alpha}, \qquad \widetilde{G}(x) \sim \widetilde{c}x^{\alpha} \qquad (x \downarrow 0), \tag{35}$$

where $c, \widetilde{c} > 0$.

Since

$$\widetilde{A}(x) = 2\widetilde{c}x^{\alpha}(1 + o(1)), \qquad \widetilde{B}(x) = O(x^{\alpha+1}),$$

identity (9) gives

$$S(x) = x - 2\widetilde{c}\,x^{\alpha+1}(1 + o(1)). \tag{36}$$

The precise recursion estimate therefore yields

$$x_n(z)^{\alpha} \sim \frac{1}{2\alpha\widetilde{c}\,n}. \tag{37}$$

Also,

$$A(x) = 2G(x) - B(x) = 2cx^{\alpha}(1 + o(1)),$$

because $B(x) = O(x^{\alpha+1})$. Hence

$$A(x_n(z)) \sim \frac{c}{\alpha\widetilde{c}}\frac{1}{n}. \tag{38}$$

Put

$$\rho := \frac{c}{\alpha\widetilde{c}}. \tag{39}$$

The logarithmic product estimate gives

$$P_n(z) = n^{-\rho+o(1)}. \tag{40}$$

Theorem 2 (Noncritical coefficient classification)**.** *Under* (35)*,*

$$\boxed{\begin{aligned} c > \alpha\widetilde{c} \quad &\Longrightarrow \quad \mathbb{E}[W] < \infty, \\ c < \alpha\widetilde{c} \quad &\Longrightarrow \quad \mathbb{E}[W] = \infty. \end{aligned}} \tag{41}$$

Proof. Suppose first that $\rho > 1$. Choose σ with $1 < \sigma < \rho$. The asymptotic relation (38), which holds uniformly for starting points in a fixed compact set, implies

$$A(x_k(z)) \geq \frac{\sigma}{k}$$

for all sufficiently large k, uniformly for $z \in I$. Since $\log(1 - u) \leq -u$,

$$P_n(z) \leq C_0 n^{-\sigma}$$

with a constant C_0 independent of $z \in I$. Thus

$$\sup_{z \in I} L(z) < \infty.$$

The same argument for the single starting point x_0 gives $L(x_0) < \infty$. The global renewal criterion and (34) now imply $\mathbb{E}[W] < \infty$.

Suppose instead that $\rho < 1$. Choose σ with $\rho < \sigma < 1$. For all sufficiently large k,

$$A(x_k(x_0)) \le \frac{\sigma}{k}.$$

The lower logarithmic estimate used in (20), together with $\sum_k A(x_k(x_0))^2 < \infty$, gives

$$P_n(x_0) \ge c_0 n^{-\sigma}$$

for some $c_0 > 0$. Hence $L(x_0) = \infty$. It follows from (32) and $x_0 > 0$ that $\mathbb{E}[W] = \infty$. □

At $c = \alpha\widetilde{c}$, the preceding proof gives $P_n(z) = n^{-1+o(1)}$, which does not determine whether $\sum_n P_n(z)$ converges. This is a genuine boundary, as the next section shows.

8 The logarithmic critical boundary

Assume that, for all sufficiently small x,

$$\widetilde{G}(x) = \widetilde{c}x^{\alpha} \tag{42}$$

and that the true mixture distribution satisfies

$$G_{\gamma}(x) = \widetilde{c}x^{\alpha}\left(\alpha + \frac{\gamma}{\log(1/x)} + O\left(\frac{1}{\log^2(1/x)}\right)\right). \tag{43}$$

Its leading coefficient is

$$c = \alpha\widetilde{c},$$

independently of γ.

The exact power form in (42) gives

$$\widetilde{B}(x) = \frac{2\widetilde{c}\alpha}{\alpha+1}x^{\alpha+1}, \qquad \widetilde{A}(x) = 2\widetilde{c}x^{\alpha} - \frac{2\widetilde{c}\alpha}{\alpha+1}x^{\alpha+1}.$$

Substitution into (9) gives

$$x - S(x) = 2\widetilde{c}x^{\alpha+1}\big(1 + O(x)\big). \tag{44}$$

Put $u_n = x_n^{-\alpha}$. From (44) and the expansion $(1-t)^{-\alpha} = 1 + \alpha t + O(t^2)$,

$$u_{n+1} - u_n = 2\alpha\widetilde{c} + O\left(x_n^{\min\{1,\alpha\}}\right). \tag{45}$$

Since $x_n = \Theta(n^{-1/\alpha})$, summing (45) shows that, for some $\delta_0 > 0$,

$$x_n^{\alpha} = \frac{1}{2\alpha\widetilde{c}n}\left(1 + O(n^{-\delta_0})\right), \tag{46}$$

where a possible $O(\log n/n)$ relative error, arising when the exponent in (45) equals 1, is absorbed by decreasing δ_0. These estimates are uniform over compact sets of starting points.

Because $B(x) \le 2xG(x)$, relation (43) implies

$$B(x_n) = O(x_n^{\alpha+1}).$$

Using $A = 2G - B$, (46), and

$$\log(1/x_n) = \frac{1}{\alpha}\log n + O(1),$$

we obtain

$$A(x_n) = \frac{1}{n} + \frac{\gamma}{n\log n} + O\left(\frac{1}{n\log^2 n}\right). \tag{47}$$

Since the squares of these terms are summable,

$$\log P_n = -\sum_{k=0}^{n-1} A(x_k) + O(1).$$

The standard sums

$$\sum_{k=2}^{n} \frac{1}{k} = \log n + O(1), \qquad \sum_{k=2}^{n} \frac{1}{k\log k} = \log\log n + O(1),$$

together with the convergence of $\sum_{k\geq 2} 1/(k\log^2 k)$, yield

$$P_n(z) \asymp \frac{1}{n(\log n)^\gamma}. \tag{48}$$

The integral substitution $u = \log t$ gives

$$\int^\infty \frac{dt}{t(\log t)^\gamma} = \int^\infty u^{-\gamma}\,du,$$

which is finite exactly when $\gamma > 1$. The corresponding positive sequence is eventually decreasing, so the integral test applies.

Theorem 3 (Logarithmic classification at the critical ratio)**.** *Within the family* (42)–(43)*,*

$$\boxed{\mathbb{E}[W] < \infty \quad\iff\quad \gamma > 1.} \tag{49}$$

Proof. By (48),

$$L(z) < \infty \quad\iff\quad \gamma > 1.$$

For $\gamma > 1$, the estimates are uniform on the reset region, so $\sup_{z\in I} L(z) < \infty$, and $L(x_0) < \infty$. The global renewal criterion then gives $\mathbb{E}[W] < \infty$.

For $\gamma \leq 1$, $L(x_0) = \infty$. Since the initial majority is wrong with probability $x_0 > 0$, the global criterion gives $\mathbb{E}[W] = \infty$. □

The family in (43) is nonempty for every $\gamma \in \mathbb{R}$. To see this explicitly, define, for sufficiently small $\varepsilon > 0$,

$$H_\gamma(x) := \widetilde{c}x^\alpha\left(\alpha + \frac{\gamma}{\log(1/x)}\right), \qquad 0 < x < \varepsilon.$$

As $x \downarrow 0$, both $H_\gamma(x)$ and its derivative are positive, because

$$H_\gamma'(x) = \widetilde{c}x^{\alpha-1}\left[\alpha\left(\alpha + \frac{\gamma}{\log(1/x)}\right) + \frac{\gamma}{\log^2(1/x)}\right].$$

Choose $\varepsilon < 1/2$ so that $H'_\gamma \geq 0$ on $(0, \varepsilon)$ and $2H_\gamma(\varepsilon) < 1$. Then the symmetric density

$$g_\gamma(q) = \begin{cases} H'_\gamma(q), & 0 < q < \varepsilon, \\ \dfrac{1 - 2H_\gamma(\varepsilon)}{1 - 2\varepsilon}, & \varepsilon \leq q \leq 1 - \varepsilon, \\ H'_\gamma(1 - q), & 1 - \varepsilon < q < 1 \end{cases}$$

is nonnegative and integrates to one. Its cumulative distribution function equals $H_\gamma(x)$ for $x < \varepsilon$, so it realizes (43) with the $O(\log^{-2}(1/x))$ remainder equal to zero. The associated conditional laws obtained from (1) are coherent and continuous.

Thus even the common exponent and both leading tail coefficients fail to decide efficiency at the critical coefficient ratio.

9 Explicit equal-exponent examples

Fix $\alpha > 0$ and $c > 0$. Choose $\varepsilon \in (0, 1/2)$ sufficiently small that

$$2c\varepsilon^\alpha < 1.$$

Define the symmetric density

$$g_c(q) = \begin{cases} c\alpha q^{\alpha-1}, & 0 < q < \varepsilon, \\ \dfrac{1 - 2c\varepsilon^\alpha}{1 - 2\varepsilon}, & \varepsilon \leq q \leq 1 - \varepsilon, \\ c\alpha(1 - q)^{\alpha-1}, & 1 - \varepsilon < q < 1. \end{cases} \tag{50}$$

The two tail pieces each have mass $c\varepsilon^\alpha$, while the middle piece has mass $1 - 2c\varepsilon^\alpha$. Thus g_c is a nonnegative probability density symmetric about $1/2$.

Let G_c be its cumulative distribution function and define

$$f_{\mathsf{h},c}(q) := 2qg_c(q), \qquad f_{\ell,c}(q) := 2(1 - q)g_c(q). \tag{51}$$

Symmetry gives

$$\int_0^1 qg_c(q)\,dq = \int_0^1 (1 - q)g_c(q)\,dq = \frac{1}{2}.$$

Hence both densities in (51) integrate to one and satisfy the calibration identities. They also have the required state symmetry.

For $x \leq \varepsilon$,

$$G_c(x) = cx^\alpha, \tag{52}$$

and direct integration gives

$$A_c(x) = 2cx^\alpha - \frac{2c\alpha}{\alpha + 1}x^{\alpha+1}, \tag{53}$$

$$B_c(x) = \frac{2c\alpha}{\alpha + 1}x^{\alpha+1}. \tag{54}$$

Use G_c as the true mixture and $G_{\widetilde{c}}$ as the perceived mixture, choosing ε small enough for both densities. In this exact-power family, (44) and (45) show more precisely that

$$x_n^\alpha = \frac{1}{2\alpha\widetilde{c}\,n} + \delta_n,$$

where

$$\sum_{n=1}^{\infty} |\delta_n| < \infty.$$

Indeed, if $\alpha > 1$, summing (45) gives

$$x_n^{-\alpha} = 2\alpha\widetilde{c}\,n + O(n^{1-1/\alpha}),$$

while if $\alpha \leq 1$, the error is $O(\log n)$. Inverting these estimates makes the displayed error summable.

Using (53), it follows that

$$A_c(x_n) = \frac{c}{\alpha\widetilde{c}}\frac{1}{n} + \epsilon_n, \qquad \sum_{n=1}^{\infty} |\epsilon_n| < \infty. \tag{55}$$

Since also $\sum_n A_c(x_n)^2 < \infty$,

$$\sum_{k=1}^{n} \left[\log\bigl(1 - A_c(x_k)\bigr) + \frac{c}{\alpha\widetilde{c}}\frac{1}{k} \right]$$

converges to a finite real number. Therefore, for every $z \in (0, 1/2]$, there is a constant $\kappa(z) \in (0, \infty)$ such that

$$P_n(z) \sim \kappa(z) n^{-c/(\alpha\widetilde{c})}. \tag{56}$$

Combining this with the global criterion gives

$$\boxed{\mathbb{E}[W] < \infty \quad \iff \quad c > \alpha\widetilde{c}} \tag{57}$$

for the explicit family. At equality, (56) is harmonic and $L(x_0) = \infty$.

Taking $\widetilde{c} = 1$ and choosing a common sufficiently small ε gives the following concrete pairs, all with the same true and perceived exponent α:

$$\begin{aligned}
c = \alpha + 1 \quad &\Longrightarrow \quad \mathbb{E}[W] < \infty, \\
c = \frac{\alpha}{2} \quad &\Longrightarrow \quad \mathbb{E}[W] = \infty, \\
c = \alpha \quad &\Longrightarrow \quad P_n(z) \sim \frac{\kappa(z)}{n} \quad \text{and} \quad \mathbb{E}[W] = \infty.
\end{aligned}$$

As a consistency check, in the well-specified specialization $c = \widetilde{c}$, the strict inequality $c > \alpha\widetilde{c}$ is equivalent to $\alpha < 1$. At $\alpha = 1$, the exact-power model lies on the harmonic boundary and is inefficient.

10 Conclusion

The equal-tail-exponent problem is completely solved under the stated continuity, symmetry, calibration, conditional independence, and two-sided power-tail hypotheses. For arbitrary, possibly oscillatory tails, the exact answer is the deterministic renewal condition (33), whose essential wrong-run component is (25). When tail constants exist away from the critical ratio, efficiency is determined by

$$c > \alpha\widetilde{c}.$$

At $c = \alpha\widetilde{c}$, logarithmic and still finer tail behavior is genuinely decisive; in the logarithmic family (43), the exact threshold is $\gamma > 1$. The explicit constructions demonstrate efficient, inefficient, and harmonic-boundary models with identical true and perceived tail exponents. No unresolved case remains within these hypotheses.

References

[1] Itai Arieli, Yakov Babichenkoyako, Stephan Müller, Farzad Pourbabaee, Omer Tamuz. *The Hazards and Benefits of Condescension in Social Learning*. Economics and Computation (EC), 2023. https://doi.org/10.1145/3580507.3597752